

Replace Zeroes with Nearest Non-Zero Value

Problem Statement

You are given an array containing zeroes and non-zero values. Replace every zero with the nearest non-zero value.

Example

Input:
[0, 0, -6]

Output:
[-6, -6, -6]

Java Solution

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        for (int i = 0; i < n; i++) {
            if (arr[i] == 0) {
                int left = i - 1;
                int right = i + 1;

                while (left >= 0 && arr[left] == 0)
                    left--;

                while (right < n && arr[right] == 0)
                    right++;

                if (left < 0)
                    arr[i] = arr[right];
                else if (right >= n)
                    arr[i] = arr[left];
                else {
                    if (i - left <= right - i)
                        arr[i] = arr[left];
                    else
                        arr[i] = arr[right];
                }
            }
        }

        System.out.println(Arrays.toString(arr));
    }
}
```